Overview of the package LMMstar

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This vignette describes the main functionalities of the **LMMstar** package.

1 S3 methods

```
print(M.class2method, quote = FALSE)
```

	lmm	Wald_lmm	rbindWald_lmm	mlmm	effect_lmm	confint_lmm
anova	x			x		
autoplot	x	x		x		
coef	x	x	x	x		
confint	x	x	x	x	x	
df.residual	x			x		
dummy.coef	x					
effects	x					
estimate	x			x		
fitted	x			x		
formula	x					
iid	x	x	x	x		
influence	x	x	x	x		
information	x					
levels	x					
lmm		x	x			
logLik	x			x		
model.frame	x					
model.matrix	x					
model.tables	x	x	x	x	x	
moments	x					
nobs	x			x		
partialCor	x					
plot	x	x		x		
pool		x	x			
predict	x			x		
print	x	x		x	x	x
profile	x					
ranef	x			x		

rbind		x		x		
resample	x				x	
residuals	x				x	
score	x				x	
sigma	x					
summary	x	x			x	x
terms	x					
variable.names	x				x	
vcov	x	x		x	x	
weights		x				

2 Likelihood and derivatives in a linear mixed model

In `lmm`:

- the argument `formula` specifies $\mu(\boldsymbol{\Theta}, \boldsymbol{X})$ the modeled mean of the outcome $\boldsymbol{Y}$ conditional a vector of p covariates $\boldsymbol{X}$.
- the argument `repetition` (typically equal to `~time|id`) enable to interpret $\boldsymbol{Y}$ as a vector of m outcomes (indexed by `time`) for which we observe independent replicates (across `id`).
- the argument `structure` define a model for the residual variance-covariance $\Omega(\boldsymbol{\Theta}, \boldsymbol{X})$.
- the argument `weights` (typically equal to `r~w`) has on its right-hand side a variable defining a vector of weights $\omega = (w_1, \ldots, w_n)$ - which should be constant within cluster - and on its left hand side a variable defining the number of replicates $\mathbf{r} = (r_1, \ldots, r_T)$ which were averaged to obtain the outcome $\boldsymbol{Y}$. When left un-specified, the weights and number of replicates are considered equal to 1.

2.1 Log-likelihood

With n independent replicates $(\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_n)$ and $(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)$, the restricted Gaussian log-likelihood is:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\Theta}|\boldsymbol{Y}, \boldsymbol{X}) =& \frac{p}{2}\log(2\pi) - \frac{1}{2}\log\left(\left|\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_i^\intercal\right|\right) \\ &+ \sum_{i=1}^{n} w_i \left(-\frac{m}{2}\log(2\pi) - \frac{1}{2}\log|\Omega_i(\boldsymbol{\Theta})| - \frac{1}{2}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal\right) \end{aligned} \tag{1}$$

where $\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta}) = \boldsymbol{Y}_i - \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)$ denotes the residuals. This is what the `logLik` method is computing for the REML criteria. The red term is specific to the REML criteria and prevents from computing individual contributions to the likelihood[1]. The blue term is what `logLik` outputs for the ML criteria when setting the argument `indiv` to `TRUE`.

Note: the implementation uses that within a subset of subject sharing the same covariance pattern:

$$\begin{aligned}\sum_i w_i \boldsymbol{X}_i \Omega(\boldsymbol{\Theta})^{-1} \boldsymbol{X}_i^\intercal &= \left(\sum_i w_i \boldsymbol{X}_i \otimes \boldsymbol{X}_i\right)^\intercal \text{vec}\left(\Omega(\boldsymbol{\Theta})^{-1}\right) \\ \sum_i w_i \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta}) \Omega(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal &= \left(\sum_i w_i \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta}) \otimes \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})\right)^\intercal \text{vec}\left(\Omega(\boldsymbol{\Theta})^{-1}\right)\end{aligned}$$

where $\otimes$ denotes the kronecker product *vec* the vectorization of a matrix. $\sum_i w_i \boldsymbol{X}_i \otimes \boldsymbol{X}_i = \sum_i \boldsymbol{X}_i^* \otimes \boldsymbol{X}_i^*$ can then be pre-computed with $\boldsymbol{X}_i^* = \sqrt{w_i}\boldsymbol{X}_i$ and does not need to be updated during the optimization process.

[1] The REML is the likelihood of the observations divided by the prior on the estimated mean parameters $\widehat{\boldsymbol{\Theta}}_\mu \sim \mathcal{N}(\mu, \left(\boldsymbol{X}\Omega^{-1}(\boldsymbol{\Theta})\boldsymbol{X}^\intercal\right)^{-1})$. This corresponds to $\frac{1}{\sqrt{2\pi}^p \left|\left(\sum_{i=1}^n \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_i^\intercal\right)^{-1}\right|} \exp\left(-(\widehat{\boldsymbol{\Theta}}_\mu - \mu)\left(2\sum_{i=1}^n \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\boldsymbol{X}_i^\intercal\right)^{-1}(\widehat{\boldsymbol{\Theta}}_\mu - \mu)^\intercal\right)$ Since μ will be estimated to be $\boldsymbol{\Theta}_\mu$, the exponential term equals 1 and thus does not contribute to the log-likelihood. One divided by the other term gives $\sqrt{2\pi}^p \left(\left|\sum_{i=1}^n \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_i^\intercal\right|\right)^{-1}$. The log of this term equals the red term.

2.2 Score

Using that $\partial \log(\det(X)) = tr(X^{-1}\partial(X))$, the score is obtained by derivating once the log-likelihood, i.e., for $\theta \in \boldsymbol{\Theta}$:

$$\mathcal{S}(\theta) = \frac{\partial \mathcal{L}(\boldsymbol{\Theta}|\boldsymbol{Y},\boldsymbol{X})}{\partial \theta} = \frac{1}{2} tr\left(\left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_j^\intercal\right)^{-1} \left(\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{X}_i^\intercal\right)\right)$$
$$+ \sum_{i=1}^{n} w_i \left(-\frac{1}{2} tr\left(\Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta}\right) + \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal + \frac{1}{2} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta}) \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal\right).$$

This is what the `score` method is computing for the REML criteria (blue and red) and the ML (blue only) criteria. When the individual contribution is requested (`indiv=TRUE`), `score` output:

$$\mathcal{S}_i(\theta) = \frac{1}{2} tr\left(\left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_j^\intercal\right)^{-1} \left(w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{X}_i^\intercal\right)\right)$$
$$+ w_i \left(-\frac{1}{2} tr\left(\Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta}\right) + \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal + \frac{1}{2} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta}) \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal\right).$$

2.3 Hessian

Derivating a second time the log-likelihood gives the hessian, $\mathcal{H}(\boldsymbol{\Theta})$, with element:

$$\mathcal{H}(\theta, \theta') = \frac{\partial^2 \mathcal{L}(\boldsymbol{\Theta}|\boldsymbol{Y},\boldsymbol{X})}{\partial \theta \partial \theta'} = \frac{\partial \mathcal{S}(\theta)}{\partial \theta'}$$
$$= \frac{1}{2} tr\left(\left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_j^\intercal\right)^{-1} \left\{\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \left(\frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial \theta \partial \theta'} - 2 \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i^{-1}(\boldsymbol{\Theta}) \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'}\right) \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{X}_i^\intercal\right.\right.$$
$$\left.\left. + \left(\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{X}_i^\intercal\right) \left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_j^\intercal\right)^{-1} \left(\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{X}_i^\intercal\right)\right\}\right)$$
$$+ \sum_{i=1}^{n} w_i \left(\frac{1}{2} tr\left(\Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} - \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial \theta \partial \theta'}\right)\right.$$
$$- \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal - \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta'}^\intercal$$
$$\left. + \frac{1}{2} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta}) \Omega_i(\boldsymbol{\Theta})^{-1} \left(\frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial \theta \partial \theta'} - \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} - \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'}\right) \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal\right).$$

where (θ, θ') are either two mean parameters, or two variance parameters, or θ is a mean parameter and θ' as variance parameter. The case θ variance parameter and θ mean parameter can be deduced by symmetry; doing so enable some simplifications since $\frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta'} \frac{\partial \Omega(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta} = 0$.

The `information` method will (by default) return the (observed) information which is the opposite of the hessian. So the previous formula shows **-1** times what the `information` method is computing for the REML criteria (blue and red) and the ML (blue only) criteria, with the default option `type.information="observed"`. When the individual contribution is requested (`indiv=TRUE`), `information`

output:

$$\mathcal{H}_i(\theta,\theta') = \frac{1}{2}tr\left(\left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta})\boldsymbol{X}_j^\intercal\right)^{-1}\left\{w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\left(\frac{\partial^2\Omega_i(\boldsymbol{\Theta})}{\partial\theta\partial\theta'} - 2\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta}\Omega_i^{-1}(\boldsymbol{\Theta})\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\right)\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{X}_i^\intercal\right.$$
$$\left.+0.5\left(w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta}\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{X}_i^\intercal\right)\left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta})\boldsymbol{X}_j^\intercal\right)^{-1}\left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta})\frac{\partial\Omega_j(\boldsymbol{\Theta})}{\partial\theta'}\Omega_j(\boldsymbol{\Theta})^{-1}\boldsymbol{X}_j^\intercal\right)\right\}\right)$$
$$+0.5\left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta})\frac{\partial\Omega_j(\boldsymbol{\Theta})}{\partial\theta}\Omega_j(\boldsymbol{\Theta})^{-1}\boldsymbol{X}_j^\intercal\right)\left(\sum_{j=1}^{n} w_j \boldsymbol{X}_j \Omega_j^{-1}(\boldsymbol{\Theta})\boldsymbol{X}_j^\intercal\right)^{-1}\left(w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{X}_i^\intercal\right)\Bigg\}\Bigg)$$
$$+ w_i\left(\frac{1}{2}tr\left(\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta} - \Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial^2\Omega_i(\boldsymbol{\Theta})}{\partial\theta\partial\theta'}\right)\right.$$
$$-\frac{\partial\mu(\boldsymbol{\Theta},\boldsymbol{X}_i)}{\partial\theta}\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal - \frac{\partial\mu(\boldsymbol{\Theta},\boldsymbol{X}_i)}{\partial\theta}\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\mu(\boldsymbol{\Theta},\boldsymbol{X}_i)}{\partial\theta'}^\intercal$$
$$\left.+\frac{1}{2}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})\Omega_i(\boldsymbol{\Theta})^{-1}\left(\frac{\partial^2\Omega_i(\boldsymbol{\Theta})}{\partial\theta\partial\theta'} - \frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta} - \frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta}\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\right)\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal\right).$$

2.4 Expected Hessian

A possible simplification is to use the expected Hessian at the Maximum (Restricted) Likelihood. Indeed for any deterministic matrix A:

- $\mathbb{E}\left[A(\boldsymbol{Y}_i - \mu(\boldsymbol{\Theta},\boldsymbol{X}_i))^\intercal|\boldsymbol{X}_i\right] = 0$
- $\mathbb{E}\left[(\boldsymbol{Y}_i - \mu(\boldsymbol{\Theta},\boldsymbol{X}_i))A(\boldsymbol{Y}_i - \mu(\boldsymbol{\Theta},\boldsymbol{X}_i))^\intercal||\boldsymbol{X}_i\right] = tr(A\mathbb{V}ar(\boldsymbol{Y}_i - \mu(\boldsymbol{\Theta},\boldsymbol{X}_i)))$

when $\mathbb{E}\left[\boldsymbol{Y}_i - \mu(\boldsymbol{\Theta},\boldsymbol{X}_i)\right] = 0$. This leads to:

$$\mathbb{E}\left[\mathcal{H}(\theta,\theta')|\boldsymbol{X}\right]$$
$$= \frac{1}{2}tr\left(\left(\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\boldsymbol{X}_i^\intercal\right)^{-1}\left\{\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\left(\frac{\partial^2\Omega_i(\boldsymbol{\Theta})}{\partial\theta\partial\theta'} - 2\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta}\Omega_i^{-1}(\boldsymbol{\Theta})\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\right)\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{X}_i^\intercal\right.\right.$$
$$\left.\left.+\left(\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta}\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{X}_i^\intercal\right)\left(\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\boldsymbol{X}_i^\intercal\right)^{-1}\left(\sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{X}_i^\intercal\right)\right\}\right)$$
$$+\sum_{i=1}^{n} w_i\left(-\frac{1}{2}tr\left(\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta}\right) - \frac{\partial\mu(\boldsymbol{\Theta},\boldsymbol{X}_i)}{\partial\theta}\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial\mu(\boldsymbol{\Theta},\boldsymbol{X}_i)}{\partial\theta'}^\intercal\right) \quad (2)$$

which is what `information` output when the argument `type.information` is set to `"expected"` (up to a factor **-1**). A similar simplification is used when `indiv` is set to `TRUE`.

2.5 Degrees of freedom

Degrees of freedom are computed using a Satterthwaite approximation, i.e. for an estimate coefficient $\widehat{\beta} \in \widehat{\boldsymbol{\Theta}}$ with standard error $\sigma_{\widehat{\beta}}$, the degree of freedom is:

$$df\left(\sigma_{\widehat{\beta}}\right) = \frac{2\sigma_{\widehat{\beta}}^4}{\mathbb{V}ar\left[\widehat{\sigma}_{\widehat{\beta}}\right]}$$

Using a first order Taylor expansion we can approximate the variance term as:

$$\begin{aligned}\mathbb{V}ar\left[\widehat{\sigma}_{\widehat{\beta}}\right] &\approx \frac{\partial \widehat{\sigma}_{\widehat{\beta}}}{\partial \boldsymbol{\Theta}} \Sigma_{\boldsymbol{\Theta}} \frac{\partial \widehat{\sigma}_{\widehat{\beta}}}{\partial \boldsymbol{\Theta}}^\intercal \\ &\approx c_\beta \left(\widehat{\mathcal{I}}_{\widehat{\boldsymbol{\Theta}}}\right)^{-1} \frac{\partial \widehat{\mathcal{I}}_{\widehat{\boldsymbol{\Theta}}}}{\partial \boldsymbol{\Theta}} \left(\widehat{\mathcal{I}}_{\widehat{\boldsymbol{\Theta}}}\right)^{-1} c_\beta{}^\intercal \Sigma_{\boldsymbol{\Theta}} c_\beta{}^\intercal \left(\widehat{\mathcal{I}}_{\widehat{\boldsymbol{\Theta}}}\right)^{-1} \frac{\partial \widehat{\mathcal{I}}_{\widehat{\boldsymbol{\Theta}}}}{\partial \boldsymbol{\Theta}}^\intercal \left(\widehat{\mathcal{I}}_{\widehat{\boldsymbol{\Theta}}}\right)^{-1} c_\beta\end{aligned}$$

where $\Sigma_{\boldsymbol{\Theta}}$ is the variance-covariance matrix of all model coefficients, $\mathcal{I}_{\boldsymbol{\Theta}}$ the information matrix for all model coefficients, c_β a matrix used to select the element relative to β in the first derivative of the information matrix, and $\frac{\partial .}{\partial \boldsymbol{\Theta}}$ denotes the vector of derivatives with respect to all model coefficients.

For the derivative of the information (i.e. **-1** times the derivative of the hessian) we consider $(\theta, \theta', , \theta'')$ are either three mean parameters, or three variance parameters, or θ is a mean parameter and θ' and θ'' are variance parameters, or θ and θ' are mean parameters and θ'' is a variance parameter. The other cases follow by symmetry; doing so enable some simplifications since $\frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta'} \frac{\partial \Omega(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta'} \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta''} = 0$. We first focus on the derivative of the ML term:

$$\begin{aligned}\frac{\partial \mathcal{H}(\theta, \theta')}{\partial \theta''} = \sum_{i=1}^n w_i \Bigg(&\frac{1}{2} tr\left(\Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta''} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial \theta \partial \theta'} - \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial^3 \Omega_i(\boldsymbol{\Theta})}{\partial \theta \partial \theta' \partial \theta''} - \Upsilon_{\theta\theta'\theta''} - \Upsilon_{\theta'\theta\theta''}\right) \\ &+ \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \left(\frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta''} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} - \frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial \theta' \partial \theta''} + \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta''}\right) \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal \\ &+ \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta''} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta'}^\intercal \\ &+ \frac{1}{2} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta}) \left(\Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial^3 \Omega_i(\boldsymbol{\Theta})}{\partial \theta \partial \theta' \partial \theta''} + \Upsilon_{\theta\theta'\theta''} + \Upsilon_{\theta\theta''\theta'} + \Upsilon_{\theta'\theta\theta''} + \Upsilon_{\theta'\theta''\theta} + \Upsilon_{\theta''\theta\theta'} + \Upsilon_{\theta''\theta'\theta}\right) \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal\Bigg).\end{aligned}$$

where

$$\begin{aligned}\Upsilon_{\theta\theta'\theta''} =& \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \left(\frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta''} - \frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial \theta' \partial \theta''}\right) \\ \Upsilon_{\theta'\theta\theta''} =& \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} \Omega_i(\boldsymbol{\Theta})^{-1} \left(\frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta''} - \frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial \theta \partial \theta''}\right) \\ \Upsilon_{\theta''\theta\theta'} =& \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta''} \Omega_i(\boldsymbol{\Theta})^{-1} \left(\frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta'} - \frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial \theta \partial \theta'}\right)\end{aligned}$$

In the special case of the expected information, the derivative of the ML term simplifies to:

$$\frac{\partial \mathbb{E}\left[\mathcal{H}(\theta, \theta') | \boldsymbol{X}\right]}{\partial \theta''} = \sum_{i=1}^n w_i \left(\frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta''} \Omega_i(\boldsymbol{\Theta})^{-1} \frac{\partial \mu(\boldsymbol{\Theta}, \boldsymbol{X}_i)}{\partial \theta'}^\intercal + \frac{1}{2} tr\left(\Upsilon_{\theta\theta'\theta''} + \Upsilon_{\theta'\theta\theta''}\right)\right)$$

The REML term can be re-expressed as:

$$\begin{aligned}
\mathcal{H}(\theta,\theta') &= \frac{1}{2} tr\left(A^{-1}\left(C_{\theta\theta'} + B_\theta A^{-1} B_{\theta'}\right)\right) \\
A &= \sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_i^\intercal \\
B_\theta &= \sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial \theta} \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{X}_i^\intercal \\
C_{\theta\theta'} &= \sum_{i=1}^{n} w_i \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta}) \left(\frac{\partial^2 \Omega_i(\boldsymbol{\Theta})}{\partial\theta\partial\theta'} - 2\frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial\theta}\Omega_i^{-1}(\boldsymbol{\Theta})\frac{\partial \Omega_i(\boldsymbol{\Theta})}{\partial\theta'}\right) \Omega_i^{-1}(\boldsymbol{\Theta}) \boldsymbol{X}_i^\intercal
\end{aligned}$$

Therefore the third derivative of the REML term $\frac{\partial \mathcal{H}(\theta,\theta')}{\partial\theta''}$ equals:

$$\frac{1}{2} tr\left(A^{-1}\left(D_{\theta\theta'\theta''} + B_{\theta''}A^{-1}C_{\theta\theta'} + C_{\theta\theta''}A^{-1}B_{\theta'} + B_\theta A^{-1}C_{\theta'\theta''} + B_{\theta''}A^{-1}B_\theta A^{-1}B_{\theta'} + B_\theta A^{-1}B_{\theta''}A^{-1}B_{\theta'}\right)\right)$$

$$\text{with } D_{\theta\theta'\theta''} = \sum_{i=1}^{n} w_i \boldsymbol{X}_i \left(\Omega_i(\boldsymbol{\Theta})^{-1}\frac{\partial^3 \Omega_i(\boldsymbol{\Theta})}{\partial\theta\partial\theta'\partial\theta''} + \Upsilon_{\theta\theta'\theta''} + \Upsilon_{\theta\theta''\theta'} + \Upsilon_{\theta'\theta\theta''} + \Upsilon_{\theta'\theta''\theta} + \Upsilon_{\theta''\theta\theta'} + \Upsilon_{\theta''\theta'\theta}\right) \Omega_i(\boldsymbol{\Theta})^{-1} \boldsymbol{X}_i^\intercal$$

2.6 Averaging over replicates

Aeraging over technical replicates $\boldsymbol{Y}^*$ to obtain $\boldsymbol{Y}$ rescales the residual variance-covariance matrix. Denote by $\Omega_i(\boldsymbol{\Theta})$ the 'single replicate' residual variance-covariance matrix which is to be specified in argument `structure` and by $\overline{\Omega}_i(\boldsymbol{\Theta})$ the variance-covariance matrix of the average replicate. Assuming independent replicates, i.e. for two replicates r and r': $\mathbb{C}ov(\boldsymbol{Y}^*_{ir}, \boldsymbol{Y}^*_{ir'}) = \Omega_i(\boldsymbol{\Theta})\mathbb{1}_{r=r'}$, we can relate $\overline{\Omega}_i(\boldsymbol{\Theta})$ to $\Omega_i(\boldsymbol{\Theta})$:

$$\overline{\Omega}_{itt'}(\Theta) = \mathbb{C}ov\left(Y_{it}, Y_{it'}\right) = \frac{1}{r_{it}r_{it'}} \sum_{r=1}^{r_{it}} \sum_{r'=1}^{r_{it'}} \mathbb{C}ov\left(Y^*_{itr}, Y^*_{it'r'}\right) = \frac{\min(r_{it}, r_{it'})}{r_{it}r_{it'}}\Omega_{itt'}(\Theta) = \frac{1}{\max(r_{it}, r_{it'})}\Omega_{itt'}(\Theta)$$

Denote by $r_{i\pi(1)}, \ldots, r_{i\pi(T)}$ the number of replicates per outcome for subject i sorted in decreasing order. Introduce:

- $\tau = \begin{bmatrix} 1 & 0 & \ldots & 0 & 0 \\ 1 & 1 & \ldots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \ldots & 1 & 0 \\ 1 & 1 & \ldots & 1 & 1 \end{bmatrix}$ the 'unit' lower triangular matrix
- $\delta(\mathbf{r}_i) = \begin{bmatrix} r_{i\pi(1)} & 0 & \ldots & 0 & 0 \\ 0 & r_{i\pi(2)} - r_{i\pi(1)} & \ldots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & r_{i\pi(T-2)} - r_{i\pi(T-1)} & 0 \\ 0 & 0 & \ldots & 0 & r_{i\pi(T-1)} - r_{i\pi(T)} \end{bmatrix}$ the diagonal matrix of the difference in number of replicates
- P the permutation matrix with element $P_{st} = \mathbb{1}_{s=\pi(t)}$

Then, denoting by $\odot$ the element-wise product (also called Hadamard product):

$$\overline{\Omega}_i(\boldsymbol{\Theta}) = \Omega_i(\boldsymbol{\Theta}) \odot (P\tau)\delta(\mathbf{r}_i)(P\tau)^\intercal$$

leading to following log-likelihood:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\Theta}|\boldsymbol{Y},\boldsymbol{X}) =& \frac{p}{2}\log(2\pi) - \frac{1}{2}\log\left(\left|\sum_{i=1}^{n} w_i \boldsymbol{X}_i \overline{\Omega}_i^{-1}(\boldsymbol{\Theta})\boldsymbol{X}_i^\intercal\right|\right) \\ &+ \sum_{i=1}^{n} w_i \left(-\frac{m}{2}\log(2\pi) - \frac{1}{2}\log\left|\overline{\Omega}_i(\boldsymbol{\Theta})\right| - \frac{1}{2}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})\overline{\Omega}_i(\boldsymbol{\Theta})^{-1}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal\right)\end{aligned}$$

The expression of the score, Hessian, degrees of freedom can be modified similarly substituting $\overline{\Omega}$ to Ω where for the derivatives one should take the element-wise product with the replication factor, e.g.:

$$\frac{\partial\overline{\Omega}_i(\boldsymbol{\Theta})}{\partial\theta} = \frac{\partial\Omega_i(\boldsymbol{\Theta})}{\partial\theta} \odot (P\tau)\delta(\mathbf{r}_i)(P\tau)^\intercal$$

Special case ($r_{it} = r_i$): number of replicates is constant within subjects across outcomes (but may vary across subjects). Then $\overline{\Omega}_i(\boldsymbol{\Theta}) = \frac{\Omega_i(\boldsymbol{\Theta})}{r_i}$. With $w_i^* = w_i r_i$, the log-likelihood becomes:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\Theta}|\boldsymbol{Y},\boldsymbol{X}) =& \frac{p}{2}\log(2\pi) - \frac{1}{2}\log\left(\left|\sum_{i=1}^{n} w_i^* \boldsymbol{X}_i \Omega_i^{-1}(\boldsymbol{\Theta})\boldsymbol{X}_i^\intercal\right|\right) \\ &+ \sum_{i=1}^{n} \left(-\frac{m}{2} w_i \log\left(\frac{2\pi}{r_i}\right) - \frac{w_i}{2}\log\left|\Omega_i(\boldsymbol{\Theta})\right| - \frac{w_i^*}{2}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})\Omega_i(\boldsymbol{\Theta})^{-1}\boldsymbol{\varepsilon}_i(\boldsymbol{\Theta})^\intercal\right)\end{aligned}$$

The expression of the score, Hessian, degrees of freedom can be modified similarly by using the modified w_i^* weight for the REML and residual terms (but not for the ML trace terms).